

Review Note: Effectiveness of Cross-Project Bug Localization

To A Eashaan Rao. From Sridhar Chimalakonda. RISHA Lab, Department of Computer Science and Engineering, IIT Tirupati. Target venue: Journal of Systems and Software. 27 July 2026.

Core Argument

This is a study, and it should read as one. The frame is: this is what we did, this is why you should believe us, this is what we found. Nothing beyond that. The technical work is complete and the design is sound. What remains is the distance between what we measured and what we currently say we showed. Several sentences reach past what the data can carry, and those are the sentences a reviewer will pick up first. A well scoped study with modest claims gets accepted. A study that argues for a principle from three models does not. Acceptance is the objective, and the fastest route to it is to bring the claims down to exactly what we measured. The title stays as it is, since the exploratory framing is the correct one for this work.

What is working

The study is good and I want to say that clearly before the list of corrections. The controlled multi-instrument protocol is a genuine contribution and I am not aware of another CPL paper that has done it. Choosing WP-small as the primary baseline is the sharpest idea in the paper, because it reframes what every prior CPL study has actually been measuring. The statistical treatment is above the norm for the venue, with paired non-parametric tests, effect sizes reported with confidence intervals, rank-biserial alongside Cohen's d, equivalence testing for every equivalence claim, and simulation-based power. Section 3 is the best written part of the paper and its design rationale is convincing. The cold-start equivalence result and the negative source-quality predictor for COOBA are both counter-intuitive and neither appears elsewhere in the literature. The hard part is done.

Where we are claiming more than we have

1. We say CPL outcomes are architecture dependent and not a property of the source to target pair. Our own Table 10 shows BLAZE and TRANSP-CNN correlated at 0.559 with p below 0.001, so the pair does carry signal. Please use: CPL outcomes vary substantially across architectures on the same source to target pairs, and practitioners may therefore need to validate CPL for their chosen model rather than relying on results reported for a different one. Same wording in the abstract, the summary box, Section 4.4, the RQ3 summary and the conclusion.
2. We have one model per paradigm, and two of the three are our own re-implementations. Please add a sentence in Section 3.2 saying that each paradigm is represented by a single model and that class level wording is shorthand for that model. Then soften the places where we write about embedding based architectures or GNN based models as categories.
3. Phrases like rules out, establishes, and can be recommended with confidence should become is consistent with, suggests, and showed consistent benefit in this corpus. The claim in the conclusion that data scarce projects no longer need to wait years is the single most exposed sentence in the paper. Please cut it.
4. The 27 to 1 asymmetry ratio has a denominator of 0.005 MRR, which is below the run to run spread of 0.07 to 0.10 that we report ourselves in Section 5.4. Please describe the shape of the distribution instead and give n for the loss median.
5. Section 5.2 says the evaluation mechanism is the variable distinguishing the three profiles, while Section 5.4 says representation quality is bundled in rather than controlled out. These cannot both stand. Keep the mechanism as an interpretation and say so plainly.

Three things to verify before anything else

These may change numbers, so please settle them before touching the prose.

1. The metrics. For TRANP-CNN, the reported MRR is higher than what the Top-1, Top-5 and Top-10 values permit, in three of the four scenarios. BLAZE and COOBA are fine. Please re-derive both MRR and Top-K from the raw rank lists and check whether they are computed over the same set of bugs. Related, Section 3.7 says MAP and MRR are identical in the single ground truth case, but they differ in every row of Table 6, and we retain patches of up to ten files, so the ground truth is not single file. The metric definitions need to be restated for the multi file case. Please do this check first, this week.
2. Win rate has two definitions in the paper. Table 7 uses gain above zero and Table 9 uses gain above 0.01. For COOBA this produces 60.3 percent in one table and 43.3 percent in the other, two pages apart, with no note. Please settle on one definition and rename the other if we want to keep it.
3. Figure 5 marks target LoC, target bug count and the LoC ratio as significant for BLAZE, while the RQ2 text says no project level feature predicts pair level gain. As written this reads as selective reporting. The finding underneath is a good one and we should lead with it: how much CPL helps depends on how few target labels the project has. That is directly on topic for a paper about data scarcity. The honest null is narrower, that similarity based features do not predict.

Two cheap additions

Both close gaps a reviewer will find, and neither needs training. The first is an untrained baseline for cold start, meaning BGE with MaxSim aggregation and no training at all, since our current comparison against FAISS confounds source training with the change in aggregation. The second is a classical IR baseline such as BM25, which is what a project with no labels actually has available, and which we ourselves describe in Section 2.1 as requiring no training data. Please also complete the ACM SIGSOFT empirical standards checklist for the General and Data Science standards and include it in the replication package. It will show us the remaining gaps before a reviewer does.

Timing

Since the experiments are done, most of what follows is writing. My estimate is one week to confirm the metrics hold and finish the derived analysis that uses only the existing results file, then two to three weeks of writing. That is about four weeks to a submittable draft and five with a review pass from me in between. Please hold the claim softening until last, after the metric check clears, otherwise you will rewrite the same sentences twice.

The core result and the protocol are clean and do not need touching. The full list of remaining items follows.

Sridhar Chimalakonda

Annexure: Remaining Revision Items

A. Conceptual

1. WP-small is 20 percent of each target's own corpus, so it ranges from about 22 examples for xarray to roughly 267 for qiskit. A project with 267 labelled bugs is not the practitioner we describe in the introduction. Report the absolute WP-small training set size per target and relate gain to absolute label count. This is also the likely explanation for the target bug count correlation in Figure 5.
2. Our central recommendation is empirical holdout validation, but a project able to hold out a validation set is not in the zero label regime the paper is written for. Acknowledge the tension in one paragraph and state roughly how many labelled bugs a usable holdout needs.
3. Architecture is doing too much work as a variable. The three models differ in input representation, ranking mechanism, adaptation mechanism, publication year and implementation provenance. Define what we mean by architecture in Section 3.2, state which dimensions co-vary, and keep causal language confined to Section 5.2 where it is labelled as interpretation.

4. The shared FAISS stage is not neutral. Retrieval uses mean pooled BGE and BLAZE reranks with MaxSim over the same BGE embeddings, so BLAZE re-scores the pool with a better aggregation of the signal that selected it, while the other two apply an independent scoring function to a pool chosen by a representation they never see. This is the strongest single explanation for BLAZE's cold start advantage and it is not currently discussed. It is also why the untrained MaxSim baseline matters.
5. The domain taxonomy has no protocol. Domain is the moderator in Table 9 and the definition of Group A. Assignment is ours with no stated procedure, no second rater and no agreement statistic. Some calls are contestable: jupyterlab is DS/ML in Table 3 but described as an IDE in Case 2, ipython is Developer Tools, prefect is workflow orchestration classified as DS/ML. State the procedure, have a second person classify independently, report agreement.
6. Group B is built around jupyterlab and is entirely cross domain, so the within versus cross domain comparison in Table 9 is partly a jupyterlab versus not jupyterlab comparison. Re-run Table 9 excluding all jupyterlab pairs and report whether the domain effect survives.
7. The domain gap score distribution is unreported. A logistic probe separating two projects' file embeddings will approach 1.0 for almost any pair. If all 63 scores sit between 0.95 and 1.0, the null correlation is range restriction rather than evidence that domain distance does not matter. Report min, median, max and a histogram.
8. State whether CP-transfer is sequential, meaning source then target, or joint. Section 5.2 speaks of initialisation and pre-training, Section 3.6 reads as joint. It changes what the condition means.
9. Section 3.6 claims that because the target budget is matched, any advantage is attributable to cross-project knowledge. Budget matching rules out extra target data but not generic effects of more training data of any kind. Either add a shuffled or mismatched source condition on the 30 Group A pairs, or soften the sentence.
10. Add a short cost paragraph in Section 5.1. CP-transfer requires training on a full source corpus, which is considerably more compute than WP-small, and a paper making deployment recommendations should say what they cost.
11. Add a small qualitative component. Ten negative transfer cases examined by hand, showing what failure looks like at the level of specific bug reports and files, would support the Why in the title better than any further statistic and needs no compute.
12. The claim that COOBA's AST representations overfit to structural patterns conflicting with jupyterlab's frontend heavy codebase is asserted without any analysis of jupyterlab's ASTs. Either run the check or mark it as a conjecture.

B. Design and Method

13. Training protocols are not comparable across models. COOBA is the only model without early stopping, and COOBA carries the divergence claim. State whether any hyperparameter search was done, with what budget, per model, and whether settings came from the original papers or were chosen by us.
14. Validate the COOBA re-implementation against its original benchmark and reported numbers. Until that exists, the negative result cannot be attributed to the adversarial paradigm rather than to our implementation. The SIGSOFT replication supplement expects this. If it is not feasible before submission, the softened claims above are what allow us to defer it, and we should have a scoped plan ready for the response letter.
15. All splits are random rather than chronological. For bug localization this leaks future information, since a test bug can have training bugs filed later touching the same file. This is not mentioned anywhere. Add it as a threat and, if feasible, run a chronological check on a few pairs.

16. Move run to run variance out of threats and into the results. A maximum WP-small spread of 0.29 MRR against mean gains of 0.018 to 0.137 belongs beside Table 6, not on page 23. Add standard deviations to Table 6.
17. The TOST margin argument is circular and its arithmetic is wrong. Rank 5 gives 0.20 and rank 4 gives 0.25, so 0.05 MRR is exactly one rank shift, not less than one. Run sensitivity at 0.02 and 0.10 and report which conclusions survive.
18. Selection criteria (ii) and (iii) are properties of the resulting set, not filters a project can individually satisfy, so the sentence about every project satisfying all three simultaneously is not well formed. Restate (i) as the filter and the other two as post hoc verification of corpus diversity.
19. Report binomial tests for the commutativity rates. BLAZE 19 of 28 is about p equal to 0.04 and COOBA 17 of 28 is about p equal to 0.17, yet we call the latter a moderate tendency.
20. Add retrieval recall at K equal to 300 per project as a column in Table 3. It determines every absolute number in the paper and currently appears as three parenthetical values inside a threats paragraph.
21. Note that the Bonferroni family of five metrics is not five independent tests, since MRR and MAP are near duplicates and the Top-K values are nested.
22. Fit the mixed effects model with target project as a random effect rather than deferring it to future work. It uses the existing results file, addresses the non-independence of the 63 pairs, and is what the architecture claim actually needs.
23. Name the COOBA and TRANP-CNN implementations explicitly in the Data Availability statement. Given the re-implementation threat they are the most important artifact in the package.
24. Note explicitly that the RQ2 null moderator results are the conclusions most exposed to the shared retrieval ceiling, since a shared bottleneck compresses variance and biases correlations toward zero.

C. Literature

25. Ben-David and colleagues on H-divergence should motivate the linear probe domain gap measure. Ganin alone is not the right grounding for that construct.
26. Section 2.3 sets up cross-project defect prediction as the analogue and then never returns to it. Our RQ2 null replicates the central conclusion of Hosseini and colleagues in a different task. Say so in the discussion as external corroboration.
27. Add the temporal evaluation literature in bug localization benchmarking, connected to item 15.
28. Verify that Zhu and colleagues 2021, TroBo, actually makes the criticism of TRANP-CNN's baseline that we attribute to it. The trailing citations on cross-model consistency, Chandramohan, Li and Reddy, do not support that clause and read as padding.

D. Numbers and Internal Consistency

29. Section 4.1.1 says the scenario ordering holds across all five metrics for BLAZE, but cold start Top-1 at 0.154 is below WP-small at 0.156.
30. Section 1 calls xarray the smallest project. It is the smallest bug corpus. docker/compose is the smallest codebase.
31. Section 5.1 item 2 says gain decreases for larger targets. Medium is 0.110 and large is 0.106, which is a plateau, not a decrease.
32. COOBA is dated 2021 twice in Section 5.2 and 2020 in Table 1 and the reference list.
33. Figure 3's caption describes three panels for three models. The figure is a single BLAZE quartile heatmap. One cell reads n equal to 0 and should be greyed and mentioned.

34. Figure 8 annotations read minus 0.32, minus 0.22 and minus 0.01, while Section 4.3.2 reads minus 0.296, minus 0.208 and minus 0.00. Regenerate from the current results file.
35. Section 5.1 item 3 gives the Hit@1 range as 15.4 to 38.5 percent. Table 11's maximum is 46.2 percent.
36. BLAZE and COOBA have an identical small target WP-small MRR of 0.219 in Table 8. It is arithmetically consistent with the Table 6 means so it is probably coincidence, but please verify.
37. COOBA Top-1 at 43.5 percent win rate and TRANSP-CNN Top-10 at 46.0 percent are marked significant with median gain of 0.000. Report tie counts in a footnote to Table 7, otherwise it reads as an error.
38. Contribution labels C1, C2 and C3 collide with the challenge labels C1, C2 and C3 on the same page. Relabel one set.
39. Section 5.4 says each target recurs approximately five times. Section 4.5 says jupyterlab has twelve WP-small runs. Give the actual per target distribution.
40. Section 5.1 item 4 says TRANSP-CNN provides the highest absolute CPL performance of the three models. Table 6 shows BLAZE ahead on MRR and far ahead on Top-1, Top-5 and Top-10. Correct to highest ceiling relative to its own within project baseline.
41. Early access citation years: Hosseini cited as 2017 but TSE volume 45 is 2019, Huo as 2019 but TSE volume 47 is 2021, Zhu as 2022 but TKDE volume 35 is 2023. Verify the bge-code-v1 release year, currently cited as 2024.
42. Mixed British and American spelling throughout, for example localization alongside regularisation and generalisation. Pick one and apply it.
43. The phrase removing the need for target annotations in these deployments appears verbatim in Sections 4.1.4, 5.1 and 6.
44. Reference list: Cooba should be COOBA, and Gomes de Oliveira Neto is sorted under O.
45. State in Section 3.6 that WP-small's 20 percent is drawn from within WP-large's 80 percent split, so the two are nested and the test set is untouched by both.
46. The criterion for practically beneficial CPL, d at least 0.5 with Bonferroni, is introduced after the results and separates 0.59 from 0.27 cleanly. Either move it to Section 3.9 as a pre-specified rule or state that it is post hoc.

E. Conclusion

47. Restructure away from the RQ by RQ recap, which repeats numbers the reader has already seen twice. Suggested order, roughly a paragraph each: what was not previously testable; what the controlled protocol shows, folding RQ1 together with the target label volume moderator since they are the same finding from two directions; what limits it, meaning the negative transfer rates, the absent metadata predictor and the variation across architectures; what follows for practice, with costs stated; what follows for the field, which is our most durable contribution and currently sits in Section 5.3 where it will be skimmed; and then future work.
48. Cut future work from four items to three. Cross language and function level are credible. The modified training objectives item is speculative and weakens the close. End on the field level recommendation rather than on future work.

F. Venue Requirements

49. Add highlights, three to five, each within 85 characters.
50. Add the CRediT author contribution statement, the declaration of competing interest and a formal funding statement.

51. Add the RISHA Lab AI assistance disclosure.

52. Fill the ORCIDs and confirm the Zenodo package contents match what the Data Availability statement lists.

Next Tasks

Order	Task
1	Re-derive MRR and Top-K from the raw rank lists for all three models and restate the metric definitions for the multi file ground truth case
2	Settle a single definition of win rate and apply it across Tables 7 and 9
3	Reframe RQ2 around target label volume and reconcile the text with Figure 5
4	Complete the derived analysis that needs no new runs: mixed effects model, domain gap distribution, Table 9 without jupyterlab, binomial tests, tie counts, standard deviations, recall column, figures regenerated
5	Run the untrained BGE MaxSim and BM25 baselines in parallel with the writing
6	Work through Annexure items A to D
7	Restructure the conclusion and add the venue requirements in Section F
8	Claim softening in a single sweep, last, once the numbers are settled
9	Complete the ACM SIGSOFT empirical standards checklist and add it to the replication package
10	Send to Ch for a review pass before submission

The original ideas, analysis, and research synthesis in this report are the intellectual contribution of the authors. AI-assisted tools were used to enhance the clarity and presentation of the writing.